COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 DATA STRUCTURE AND ALGORITHM	**SECTION:**	CPE21S2
DATE PERFORMED:	July 10, 2026	**DATE SUBMITTED:**	July 15, 2026
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ACTIVITY NO. 1.1 **Hands-On Activity 1.1 Basic C++ Programming**

1. PROCEDURE OUTPUT

SUPPLIMENTARY

[*] 1 SWAPPING.cpp

```cpp
#include <iostream>
using namespace std;

int main() {

    int num1, num2, temp;

    cout << "Enter first number: ";
    cin >> num1;

    cout << "Enter second number: ";
    cin >> num2;

    cout << "\nBefore swapping:" << endl;
    cout << "num1 = " << num1 << endl;
    cout << "num2 = " << num2 << endl;

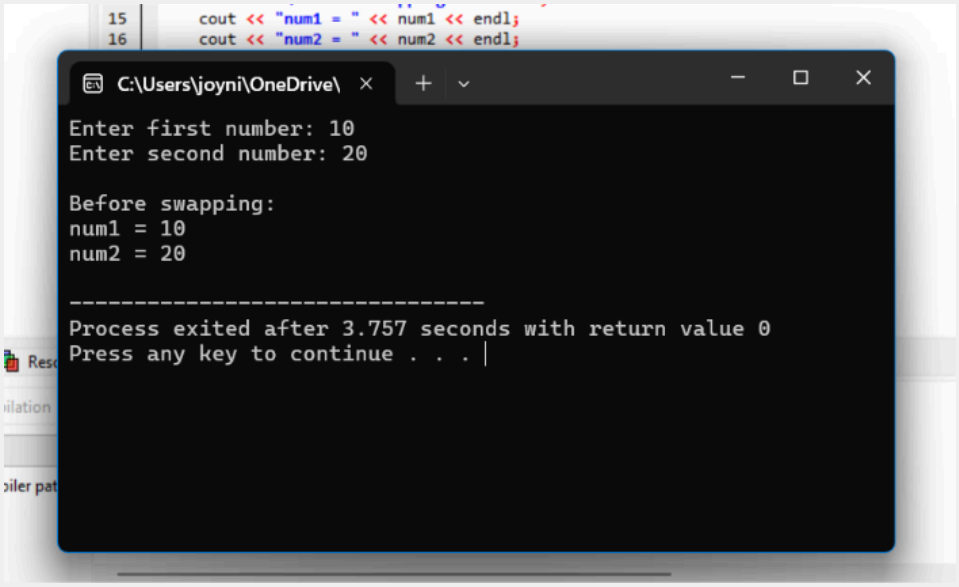
    // Swap the values

    temp = num1;
    num1 = num2;
    num2 = num3;

    cout << "\nAfter swapping:" << endl;
    cout << "num1 = " << num1 << endl;
    cout << "num2 = " << num2 << endl;

    return 0;

}
```


15 cout << "num1 = " << num1 << endl;
16 cout << "num2 = " << num2 << endl;
C:\Users\joyni\OneDrive\
Enter first number: 10
Enter second number: 20
Before swapping:
num1 = 10
num2 = 20

Process exited after 3.757 seconds with return value 0
Press any key to continue . . .

CODE 2

[*] 1 SWAPPING.cpp | ILOB.cpp | KELVIN TO FAHRENHEIT.cpp

```cpp
#include<iostream>
#include <cmath>
using namespace std;


double convertKelvinToFahrenheit(double kelvin) {
    return (kelvin - 273.15) * 9/5 + 32;
}

double distance(double x1, double y1, double x2, double y2) {
    return sqrt(pow(x2 - x1, 2) + pow(y2 - y1, 2));
}

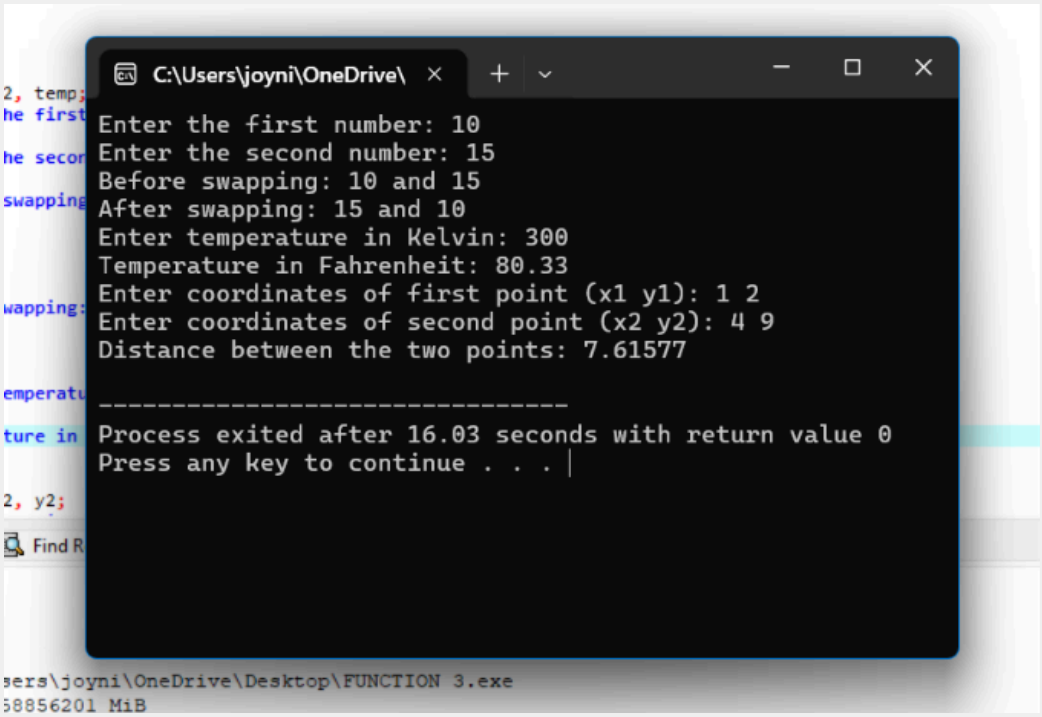
int main(){

    double num1, num2, temp;
    cout << "Enter the first number: ";
    cin >> num1;
    cout << "Enter the second number: ";
    cin >> num2;
    cout << "Before swapping: " << num1 << " and " << num2 << endl;

    temp = num1;
    num1 = num2;
    num2 = temp;
    cout << "After swapping: " << num1 << " and " << num2 << endl;


    double kelvin;
    cout << "Enter temperature in Kelvin: ";
    cin >> kelvin;
    cout << "Temperature in Fahrenheit: " << convertKelvinToFahrenheit(kelvin) << endl;


    double x1, y1, x2, y2;
    cout << "Enter coordinates of first point (x1 y1): ";
    cin >> x1 >> y1;
    cout << "Enter coordinates of second point (x2 y2): ";
    cin >> x2 >> y2;
    cout << "Distance between the two points: " << distance(x1, y1, x2, y2) << endl;
    return 0;
}
```


C:\Users\joyni\OneDrive\
Enter the first number: 10
Enter the second number: 15
Before swapping: 10 and 15
After swapping: 15 and 10
Enter temperature in Kelvin: 300
Temperature in Fahrenheit: 80.33
Enter coordinates of first point (x1 y1): 1 2
Enter coordinates of second point (x2 y2): 4 9
Distance between the two points: 7.61577
Process exited after 16.03 seconds with return value 0
Press any key to continue . . .

CODE 4

SUP4.cpp

```cpp
#include <iostream>
class Triangle {
private:
    double totalAngle, angleA, angleB, angleC;
public:
    Triangle(double A, double B, double C);
    void setAngles(double A, double B, double C);
    const bool validateTriangle();
};
// Constructor definition to initialize private member variables and calculate total angle
Triangle::Triangle(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A+B+C;
}
// Setter method definition to update angle values and recalculate the overall sum
void Triangle::setAngles(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A+B+C;
}
// Method to determine if overall angle is triangle or not
const bool Triangle::validateTriangle() {
    return (totalAngle <= 180);
}
// Driver code
int main(){
    Triangle set1(40, 30, 110);
    if(set1.validateTriangle()){

    std:: cout << "The shape is a valid triangle.\n";

    } else {
      std:: cout << "The shape is NOT a valid triangle.\n";
    }
    return 0;
}

```

```
C:\Users\TIPQC\OneDrive\De:
The shape is a valid triangle.

--------------------------------
Process exited after 0.03798 seconds with return value 0
Press any key to continue . . .
```

SUP4.cpp | Function Supplimentary 3.cpp

```cpp
#include <iostream>
#include <cmath>

using namespace std;

// Function to calculate distance
double calculateDistance(double x1, double y1, double x2, double y2) {
    return sqrt(pow(x2 - x1, 2) + pow(y2 - y1, 2));
}

int main() {
    double x1, y1, x2, y2;

    cout << "Enter x1: ";
    cin >> x1;

    cout << "Enter y1: ";
    cin >> y1;

    cout << "Enter x2: ";
    cin >> x2;

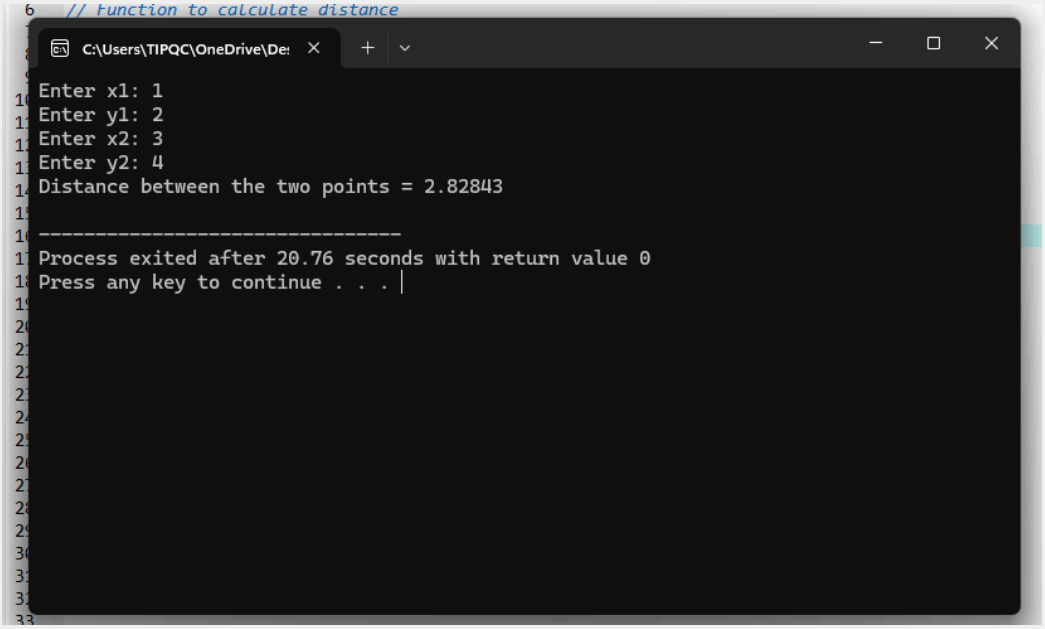
    cout << "Enter y2: ";
    cin >> y2;

    double distance = calculateDistance(x1, y1, x2, y2);

    cout << "Distance between the two points = "
         << distance << endl;

    return 0;
}

```



What procedures did you perform?

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: Gemini, Google and grammarly

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: Gemini I used that to assist and explain my code , Google for translation, the grammarly to correct my grammar

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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